

Q1. Discuss ASME code for shaft design.

One important approach of designing a transmission shaft is to use the ASME code. According to this code, the permissible shear stress τ_{max} for the shaft without keyways is taken at 30% of yield strength to tension or 18% of the ultimate tensile strength of the material, whichever is minimum. Therefore,

$$\tau_{max} = 0.30 S_{yt}$$

$$\text{or } \tau_{max} = 0.18 S_{ut} \quad \text{--- ①}$$

If keyways are present, the above values are to code the bending & torsional moments are to be multiplied by factors k_b & k_t respectively to account for shock & fatigue in operating condition. The ASME code is based maximum shear stress theory of failure. Therefore,

$$\tau_{max} = \frac{16}{\pi d^3} \sqrt{(k_b M_b)^2 + (k_t M_t)^2} \quad \text{--- ②}$$

Where,

k_b = combined shock & fatigue factor applied to bending moment.

k_t = Combined shock & fatigue factor applied to torsional moment.

The values of k_b and k_t for rotating shaft are given table.

Application	k_b	k_t
i. load gradually applied	1.5	1.0
ii. load suddenly applied (minor shock)	1.5 - 2.0	1.0 - 1.5
iii. load suddenly applied (heavy shock)	2.0 - 3.0	1.5 - 3.0

Eqⁿ ① and ② are used to design the shaft according to the ASME code.

$$\tau_{\max} = \frac{16}{\pi d^3} \sqrt{(k_b M_b)^2 + (k_t M_t)^2}$$

$$\sigma = \frac{16}{\pi d^3} \left[k_b M_b + \sqrt{(k_b M_b)^2 + (k_t M_t)^2} \right]$$

1.] Equivalent Torsional Moment.

The expression torsional moment when the shaft is subjected to fluctuating loads the equivalent torsional moment is defined as the torsional moment, which when acting alone, will produce the same torsional stress in the shaft as under the combined action of bending moment (M_b) and torsional moment (M_t) under fluctuating loads.

2.] Equivalent Bending Moment.

The expression $\sqrt{(k_b M_b)^2 + (k_t M_t)^2}$ is called equivalent torsional moment when the shaft is subjected to fluctuating load. The same torsional stress is the shaft as under the combined action of bending moment (M_b) and torsional moment (M_t) under fluctuating loads.

Q.2 What do you understand by transmission shaft and splined shaft [lined shaft] Differential betⁿ Axle and spindles.

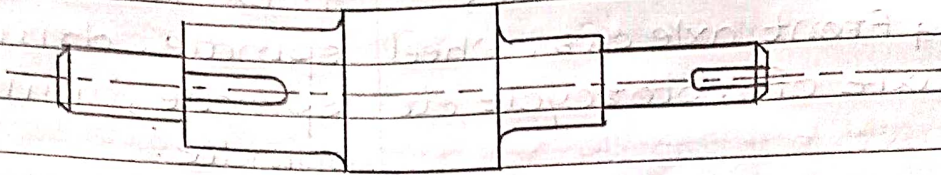


fig. Transmission Shaft.

The term 'transmission shaft' usually refers to a rotating machine element circular in cross-section, which supports transmission elements like gears, pulleys & sprockets and transmits power. A transmission shaft supporting a gear in a speed reducer is shown in fig. The shaft is always stepped with maximum diameter in the middle portion & minimum diameter at two ends. The rounded-off portion betⁿ two cross-sections of different diameters is called fillet. The fillet radius is provided to reduce the effect of stress-concentration due to abrupt change in the cross-section.

* Differential betⁿ Axle and spindles.

Axle	Spindles
1. Axle is non-rotating member	1. spindle is a rotating member.
2. The purpose of a axle is only to support the transmitting elements	2. The spindle is a short rotating shaft which is an integral part of the machine.

3. Axles is subjected to bending moment and axial force

3. A Spindle is subjected to bending moment and axial force.

4. Ex. lathe machine, Front axle car, wheel axle of motorcycle etc.

4. Ex. lathe machine spindle, drilling m/c spindle, Milling machine spindle.

Q3 What are the diff. types of keys used (Classification).

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Keys -

It is defined as a machine element, which is used to connect the transmission shaft to rotating machine elements like pulley, gear, sprocket or flywheel.

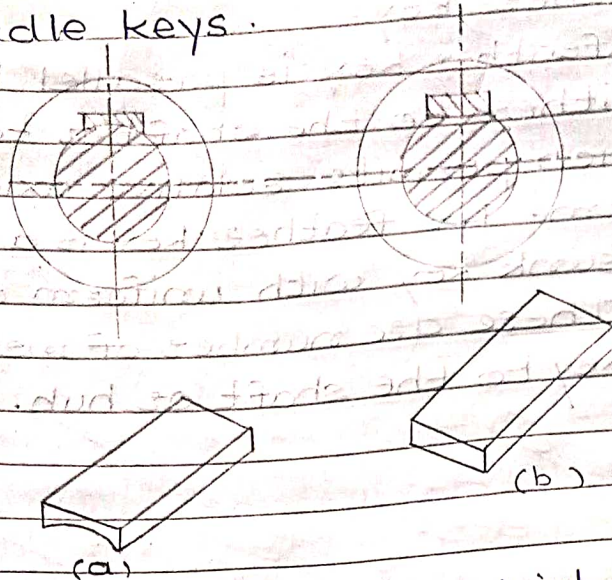
Many types of key are available and these are numbers of standards, which specify the dimensions of the key.

There are different ways to classify the keys.

- a) Saddle key and sunk key
- b) Square key and flat key
- c) Taper key and parallel key
- d) key with and without Gib-head.

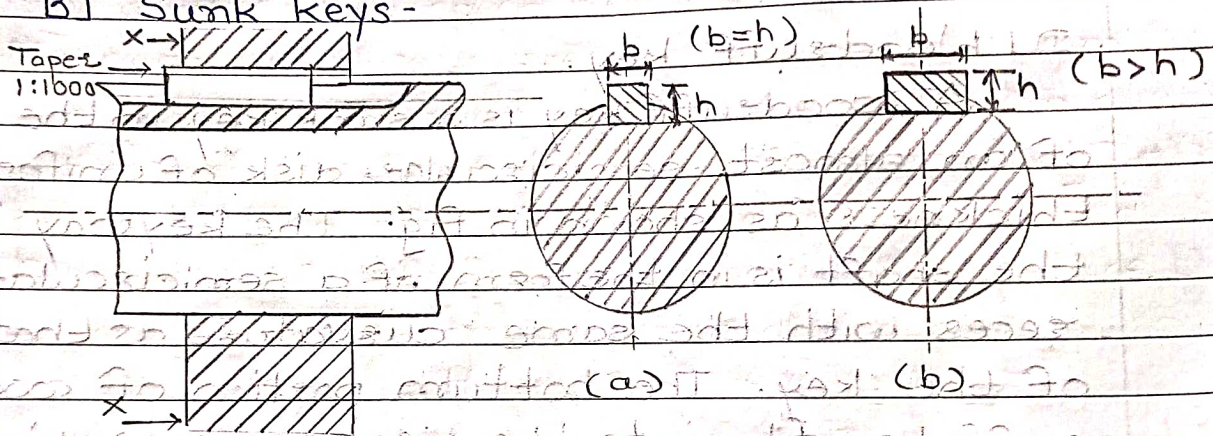
In addition, there are special types of keys such as woodruff key, Kennedy key, or feather key.

A1 Saddle keys.



A saddle key is a key, which fits in the keyways of the hub only. In this case there is no keyways on the shaft. There are two types of saddle key, namely, hollow and flat.

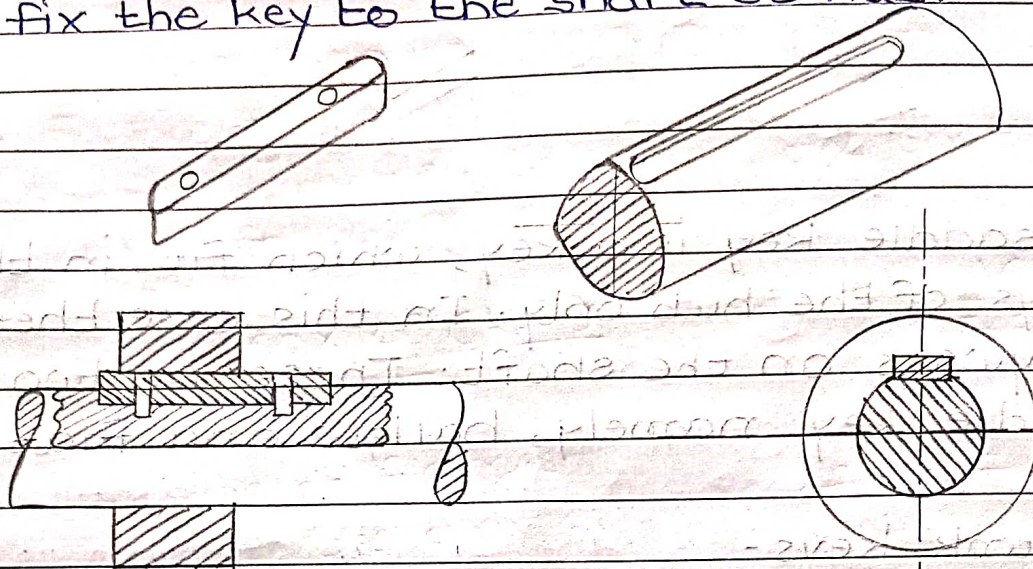
B1 Sunk keys -



A sunk key is a key which, half the thickness of the key fits into the keyway on the shaft & the remaining half in the keyway on the hub. Sunk key is suitable for heavy duty applications, since there is no possibility of the key to slip around the shaft. It is a positive drive. This is the main advantage of the sunk key over the saddle key.

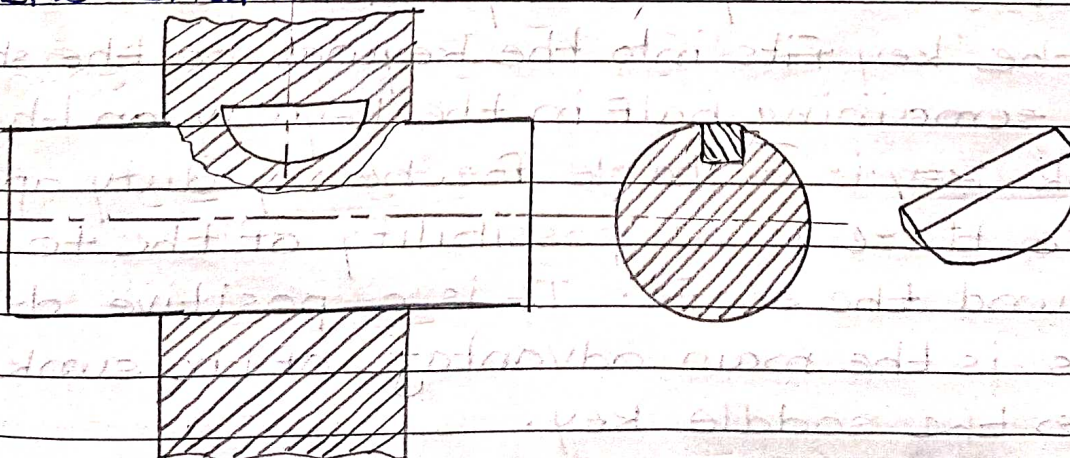
C1 Feather Key.

A feather key is parallel key which is fixed either to the shaft or to the hub and which permits relative axial movement between them. The feather key is a particular type of sunk key with uniform width and height. There are number of methods to fix the key to the shaft or hub.



D1 Woodruff Key.

A woodruff key is a sunk key in the form of an almost semicircular disk of uniform thickness as shown in fig. The keyway in the shaft is in the form of a semicircular recess with the same curvature as that of the key. The bottom portion of woodruff key fits into the circular keyway in the shaft.



Q4. What are diff. type of Couplings.
 A) Discuss design procedure for muff coupling
 B) Discuss design procedure for rigid flange coupling.

→ A) Design procedure for muff coupling.
 The basic procedure for finding out the dimensions of the muff coupling consist of the following steps.

i) Calculate the diameter of each shaft by the following eqn:

$$M_t = \frac{50 \times 10^6 \text{ (kW)}}{2\pi n} \quad \text{and} \quad \tau = \frac{16 M_t}{\pi d^3}$$

ii) Calculate the dimensions of the sleeve by the following empirical eqn.

$$D = (2d + 13) \text{ mm} \quad \text{and} \quad L = 3.5d$$

Also check the torsional shear stress induced in the sleeve by the following eqn.

$$\tau = \frac{M_t \cdot \varepsilon}{J}, \quad J = \frac{\pi (D^4 - d^4)}{32}, \quad \varepsilon = \frac{D}{2}$$

iii) Determine the standard cross-section of the sunk key. The length of the key in each shaft is one-half of the length of sleeve.

$$l = \frac{L}{2}$$

With these dimensions of the key, check the shear & compressive stress in the key by eqn.

$$\tau = \frac{2 M_t}{d b l} \quad \text{and} \quad \sigma_c = \frac{4 M_t}{d h l}$$

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The shaft and key are made of plain carbon steel. The sleeve is usually made of gray cast iron of grade FG 200.

B1 Discuss design procedure for rigid Flange couplings.

→ The basic procedure for finding out the dimensions of the rigid flange coupling consist of following steps:

- i) Shaft diameter - Calculate the shaft diameter by using following two eqn.

$$M_t = \frac{60 \times 10^6 (kW)}{2\pi n} \quad \text{and} \quad \tau = \frac{16 M_t}{\pi d^3}$$

- ii) Dimensions of flanges -

calculate the dimensions of the flange by follows empirical eqn.

$$d_h = 2d, \quad h_f = 1.5d, \quad D = 3d;$$

$$t = 0.5d, \quad t_f = 0.25d$$

$$d_s = 1.5d, \quad D_o = (4d + 2t_f)$$

The torsional shear stress in the hub can be calculated by considering it as a hollow shaft subjected to torsional moment M_t .

The inner and outer diameters of hub are d and d_h respectively. The torsional shear stress in the hub is given by.

$$\tau = \frac{M_t \cdot r}{J} ; \quad J = \frac{\pi (d_h^4 - d^4)}{32} \quad \text{--- ①}$$

$$r = \frac{d_h}{2}$$

$$\text{Area under shear} = (\pi d_h) \times t.$$

$$\text{Shear force} = \text{Area} \times \text{Shear stress}$$

$$= \pi d_h t \tau$$

$$\text{resisting torque} = \text{force} \times \left(\frac{d_h}{2}\right)$$

$$M_t = (\pi d_h t \tau) \times \frac{d_h}{2}$$

$$M_t = \frac{1}{2} \pi d_h^2 t \tau \quad \text{--- (2)}$$

iii) Diameter of bolts -

Decide the number of bolts using following guidelines-

$$N = 3 \text{ for } d < 40 \text{ mm}$$

$$N = 4 \text{ for } 40 \leq d < 100 \text{ mm}$$

$$N = 6 \text{ for } 100 \leq d < 180 \text{ mm}$$

Determine the diameter of bolt by eqⁿ
rearranging the eqⁿ.

$$d_t^2 = \frac{8 M_t}{\pi D N \tau} \quad \text{--- (3)}$$

Where, τ is permissible shear stress for the bolt material.

$$\text{Crushing area of each bolt} = d_t t$$

$$\text{Crushing area of all bolt} = N d_t t$$

$$\text{Compressive force} = N d_t t \sigma_c$$

$$\sigma_c = \frac{2 M_t}{N d_t t D} \quad \text{--- (4)}$$

ix) Dimensions of key -

The length of the key in each shaft is l_k ,
 $\therefore l = l_k$.

With these dimensions of key, check the shear and compressive stress in the key.

$$\tau = \frac{2 M_t}{d b l} \quad \text{or} \quad \sigma_c = \frac{4 M_t}{d h l}$$